

--	--	--	--	--	--	--	--	--	--

B. Tech. Degree V Semester Examination November 2014

ME 1506 POWER PLANT ENGINEERING

(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Define depreciation of power plants and explain any method for calculating annual depreciation.
- (b) Write notes on power house equipments.
- (c) Illustrate the advantages of hydroelectric power plants.
- (d) List out the advantages of gas turbine over steam turbine.
- (e) Derive an expression to find chimney diameter at any cross section of the chimney.
- (f) List the safety measures for nuclear power plants.
- (g) Explain 'solar thermal collection'.
- (h) Describe H_2O_2 fuel cells, with neat sketch.

PART B

(4 x 15 = 60)

- II. (a) A power station is to supply three region of load whose peak loads are 20MW, 15MW and 25 MW. The annual load factor is 50% and the diversity factor of the load at station is 1.5. Determine the following: (7)
- (i) Maximum demand on the station
- (ii) Installed capacity suggesting number of unit
- (iii) Annual energy supplied.
- (b) A base load power station and standby power station share a common load as follows: (8)

Base load station annual output	=	$150 \times 10^6 \text{ kWh}$
Base load station capacity	=	35MW
Maximum demand on base load station	=	30MW
Standby station capacity	=	18MW
Standby station annual output	=	$14 \times 10^6 \text{ kWh}$
Maximum demand (peak load) on standby station	=	15MW

Determine the following parameters for both power stations.

- (i) Load factor (ii) Capacity factor (Plant factor)

OR

- III. (a) Describe four types of tariffs for electricity. (8)
- (b) With a neat sketch, make a note on load curve. Describe different types of load curves. (7)

(P.T.O.)

- IV. Explain different methods for improving thermal efficiency of open cycle gas turbine power plant, with neat sketches. (15)

OR

- V. (a) Explain the layout of diesel engine power plant, with a neat sketch. (7)
 (b) Explain the functions of following systems in a diesel engine power plant
 (i) Fuel system (4)
 (ii) Cooling system (2)
 (iii) Lubricating system (2)

- VI. (a) Describe the fluidized bed combustion system, with neat sketch. (7)
 (b) Explain dust and ash handling system, with neat sketch. (8)

OR

- VII. (a) A steam power plant uses coal 5000kg/hr. The heat conversion efficiency is 30% and calorific value of coal is 7000 k.cal/kg. Calculate the electric energy produced per day. (1kWh=869 k.cal). (7)
 (b) Explain BWR and PWR in a nuclear power plant, with neat sketches. (8)

- VIII. (a) Distinguish between thermoelectric conversion and thermionic conversion system. (8)
 (b) Describe Ocean Thermal Energy Conversion (OTEC). (7)

OR

- IX. (a) Explain MHD power generation, with neat sketches. (7)
 (b) Explain different types of solar collectors, with neat sketches. (8)
